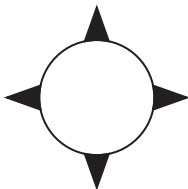
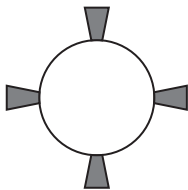
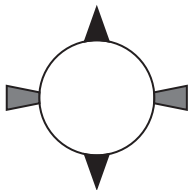
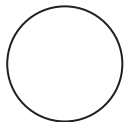


8 Humans can have one of four different blood types: A, B, AB or O.

Blood type is determined by antigens present on the membranes of red blood cells.

The table shows which antigens are present in each blood type.

Blood type	Antigens on the membrane of red blood cells	Diagram of one red blood cell
A	antigen A only	
B	antigen B only	
AB	both antigens A and B	
O	neither antigen A nor B	

Blood transfusions are used in the treatment of some diseases.

A blood transfusion involves taking blood from a healthy person and putting it into the person needing the treatment.

(a) (i) Which blood type can be used for all blood transfusions?

(1)

- ☒ **A** blood type A
- ☒ **B** blood type B
- ☒ **C** blood type AB
- ☒ **D** blood type O

(ii) Explain what will happen if a person with blood type A is given a transfusion of blood type AB.

(4)

(b) Bacteria live in our intestines (gut flora).

The membranes of cells lining our intestines have molecules similar to the antigens on red blood cells.

(i) Explain why bacteria living in our intestines are important.

(2)

- (ii) Both the molecules on the membranes of the cells lining our intestines and the antigens on red blood cells have sugars attached to protein molecules.

Bacteria living in our intestines secrete an enzyme that separates the sugars from the protein molecules.

Suggest why bacteria living in our intestines secrete this enzyme.

(2)

- (iii) Explain why this enzyme may be useful in blood transfusions.

(2)

(Total for Question 8 = 11 marks)